Characteristics of Synthetic Fibres

ACTIVITY 5

Burning A Piece of natural fibre and a synthetic Fibre

Note: This activity should be performedonly by the teacher.

Take a piece of cotton cloth and a piece of polyester cloth. Both the pieces of cloth can be of same size (2cm x 2cm is enough). Hold the pieces of cloth using tongsto protect yourself from the flame and heat. Burn both the pieces of cloth one by one and see what happens when they burn.

What do you observe while the cotton cloth burns?

……

What do you observe while the polyester cloth burns?

……

You must have noticed that natural (cotton) fibresburn very differentlycompared tosynthetic (polyester) fibres.

The cotton cloth will burn when it is held in flame. On the other hand, the synthetic polyester cloth will melt on burning. This is one of the disadvantages of wearing clothes madeout of synthetic fibres. If a synthetic cloth comes into contact with fire the fabric melts and sticks to the body causing severe burns. Therefore, one should not wear synthetic clothes while cooking or working in a laboratory.



Cotton Laboratory Coat

ACTIVITY 6

Wetting A Cotton Cloth And An Umbrella Cloth

We use an umbrella on rainy days. What kind of umbrella do we use?

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Can you use an umbrella made of cotton?

Let us do an activity to see why we do not use a cotton umbrella to protect ourselves from the rain.

Take a piece of cotton cloth of approximately 10 cm X 10 cm size and a piece of nylon or polyester cloth 10 cm X 10 cm in size from an old unusable / untorn umbrella. Ask four students to hold the four corners of the piece of cotton cloth and pour a glass of water over it. Then ask four students to hold the piece of umbrella cloth and pour a glass of water over it. Compare the effect of water on the piece of cotton cloth and umbrella cloth and record your observation in the notebook.

……

Which of these fabrics allows water to pass through?

Cotton cloth or Umbrella cloth (nylon or polyester) Now ask the students to put both the pieces of cloth in the hot sun to dry. Which of these fabrics dries the fastest? The cotton cloth or the umbrella cloth? ……...

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